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Civil & Structural Engineer. Precast Engineering Lead, Digital Transformation, SAP S/4HANA, BIM, Engineering Systems

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Profile

Civil and structural engineer with technical ownership of precast design, tendering, submittals and drawings at a Saudi precast concrete manufacturer. Led its SAP S/4HANA Cloud master data and cross-department integration through Go-Live on 30 Dec 2025, connecting engineering with costing, procurement, production and planning, and led the BIM transformation of its design office; trained the technical office and users across departments. Designs RC pipes, jacking pipes, circular and rectangular manholes and box culverts to AASHTO LRFD, ASTM, BS EN 1916, DWA-A 161, ACI and MOMRA with SAP2000 finite-element models, and has programmed those codes into design engines that check every configuration before release. Priced ~4,000 quotations since 2023 across Saudi giga- and mega-projects. Architected the Precast Engineering ERP and builds these engineering systems by directing AI coding agents under automated test gates.

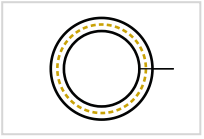
The precast order, digitalised

The Precast Engineering ERP organises each order around one inquiry record, from the client's BOQ to the SAP product, with the design, drawing and submittal engines linked into the same record.

Inquiry & pricing	Design	Drawings & BIM	Costing	SAP product	Production
~4,000	**5**	**6,394**	**TABS**	**~40,000**	**88**
quotations since 2023, across Saudi giga- and mega-projects	product families with design engines to ASTM, AASHTO and ACI	Revit sheets for 1,345 manhole products	activity-based costing used with MAS	materials, 34,000+ BOMs and ~11,000 routings	work centers tying routings to costing and planning

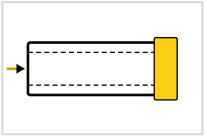
Product families

Design, drawings and BIM are automated for five precast product families.



RC pipes

78-product standard range designed to ASTM C76 and AASHTO direct design; 67 standard drawing sets.



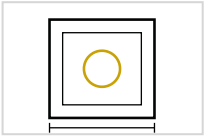
Jacking pipes

Design engine, calculation memos, catalogue and Revit drawing sets.



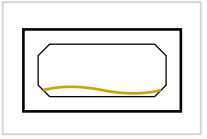
Circular manholes

Revit manhole product system v1.0, with a 6,394-sheet standard library for 1,345 manhole products.



Rectangular manholes

SAP2000 design engine; analysis cycle cut from 28+ minutes to about 6.



Box culverts

Single and twin cell designer v2.0: memos, BBS and drawings under 7,000+ tests.

Structural design

Each product family has its own design engine with these codes programmed in, so every configuration is checked before a memo, drawing or bill of materials is released.

RC pipes

Codes: ASTM C76M, C497; AASHTO LRFD 2024 §12.10; ACPA; MOMRA.

Loads: Earth (vertical arching), fluid, HL-93 and MOMRA vehicles, dry and submerged.

Analysis: Indirect design (installation Types 1-4); direct Heger design on a closed ring; Eriksson Pipe and SAP2000 cross-checks.

Checks: Flexure, crack control, diagonal tension, radial tension, steel limits, stirrups, flotation.

Jacking pipes

Codes: BS EN 1916, BS 5911-1, DWA-A 161, ATV-DVWK-A 127, EN 1992-1-1.

Loads: Earth by soil group, groundwater, traffic, jacking force with joint deflection.

Analysis: DWA-A 161 ring; packer stiffness; SAP2000 ring frame and 3D solid end-zone model.

Checks: Ring flexure and crack width, fatigue, end-zone splitting and padding ties, collar and packer.

Manholes

Codes: AASHTO LRFD 10th, ASTM C478M, ACI 318M-25, ACI 350, MOMRA live load.

Loads: Earth (ACPA arching), at-rest pressure, HL-93 with surcharge, groundwater and buoyancy, handling.

Analysis: SAP2000 shell model per piece on compression-only joints and springs; Wood-Armer slab design.

Checks: Biaxial flexure, hoop forces, shear, strut-and-tie and punching at openings, cracking, flotation, bearing.

Box culverts

Codes: AASHTO LRFD 10th or MOMRA, ASTM C1433M / C1577, ACI 318M-25.

Loads: Earth (Marston and trench factors), lateral earth, surcharge, water, HL-93, HS20, MOMRA vehicles.

Analysis: SAP2000 frame on compression-only Winkler springs; every vehicle placement evaluated.

Checks: Flexure with thrust, shear, crack control, minimum steel, wall stability, bearing, flotation, lifting.

Steel moulds

Codes: BS EN 10025 (S275JR), EN 1993-1-8, BS EN ISO 3834-3, ISO 5817, BS EN 1916 tolerances.

Loads: Fresh-concrete pressure as a full liquid head with a factor of 1.5 for vibration; lifting.

Analysis: Part-by-part checks inside generated Revit mould sets.

Checks: Lifting lugs, governing ratio per part, product and mould tolerances, shop checks before the first cast.

Experience

Senior Technical Engineer & SAP Master Data Coordinator

Aug 2023 – Present

Al Jazirah Development Factory for Concrete Products (JDCO), Riyadh, Saudi Arabia

Precast concrete manufacturer of pipes, manholes and box culverts for infrastructure. Technical owner of precast design, tendering, submittals and drawings. As the company's digital transformation lead, led the SAP S/4HANA Cloud master data and cross-department integration through Go-Live and the BIM transformation of the design office; connected engineering with costing, procurement, production and planning, and trained the technical office and users across departments.

Structural Design & Precast Standards

- Designed out human error: encoded the governing codes (ASTM, AASHTO, ACI) and programmed every configuration of the factory's precast range, so each design, drawing and bill of materials is checked against code before release.
- Standardised the RCP range: 78 products (26 moulds × Classes III/IV/V) to ASTM C76 and AASHTO direct design, with calculation memos, 67 standard drawing sets from a Revit pipeline, bar bending schedules and an inspection and test plan; the design engine's cross-check against Eriksson Pipe agreed within 0.04%.
- Delivered a SAP2000-driven design engine for circular and rectangular manholes that produces 44–54-page calculation memos; all 4 release cases passed with zero failed checks, and the rectangular analysis cycle fell from 28+ minutes to about 6.
- Took the box-culvert design program (single and twin cell, SAP2000 API) to v2.0, generating calculation memos, BBS and shop drawings under a 7,000+ test release gate.
- Built the jacking-pipe standard for microtunnelling: a 21-mould range with two cages, 50 and 100-year design lives, lined and unlined, designed to BS EN 1916 and DWA-A 161 and verified in SAP2000 (ring frame, 3D solid end zone, intermediate jacking station), with calculation memos, a catalogue and Revit drawing sets per size.
- Designed the steel moulds for the 21-size jacking-pipe range, with a full Revit drawing set per size: pallet, jacket, collapsible core, header and intermediate jacking station units, checked for fresh-concrete pressure under vibration and for lifting to EN 1993-1-8, with tolerances from BS EN 1916 and welding to BS EN ISO 3834-3.
- Structured a revision-controlled registry of Saudi authority standards and standard drawings for precast products, feeding the company's product-standards matrix.

Technical Office: Tendering, Submittals & Approvals

- Technical owner of quotation engineering: priced ~4,000 quotations since 2023 across Saudi giga- and mega-projects, including NEOM, Qiddiya, Diriyah, King Salman Park, EXPO 2030, King Salman International Airport, KAFD, Salwa and Royal Commission work: BOQ and item analysis, product design where needed, quantity take-off, costing, drawings and, since Go-Live, SAP material creation.
- Built a gated BOQ take-off pipeline for tender packages that generates SAP material codes directly from BOQ attributes and turns every error found into a reusable check.
- Produced the engineering documents for technical review and submittals: calculation memos from the design engines (156 for the RC pipe range alone), an 86-page engineering catalogue and a 35-page bilingual public catalogue, standard Revit drawing sets with BBS, and an inspection and test plan specifying three-edge-bearing proof testing to ASTM C497.
- Prepared prequalification and material-submittal packages for contractor and consultant review on projects including Qiddiya, EXPO 2030, King Salman Park, Diriyah and KAFD, logging review outcomes to fix their root causes.
- Built the prequalification and material-submittal system: an 18-category reference library, a certificate-expiry register and compliance sheets per product; rebuilt the 452-page master prequalification package (156 bookmarks) in the company identity.

Manufacturing, Costing & Cross-Department Integration

- Connected engineering to costing, procurement, production and planning through SAP product structures (BOMs, routings, work centers linked to cost centers) that drive material requirements, purchasing, production orders and TABS costing.
- Modelled the factory's production resources in SAP: 88 work centers with 337 cost-center links, 10,931 production versions and routings with 41,000+ operations, so capacity, costing and planning run on the same engineering data.
- Developed the TABS activity-based costing model (direct material; production overhead, labour and machine; transport) and a pricing engine with floor/target/max logic per product family, used alongside MAS for quotation pricing.

SAP S/4HANA Cloud Digital Transformation

- Led master data and cross-department integration for JDCO's SAP S/4HANA Cloud (Public Edition) implementation through Go-Live on 30 Dec 2025, connecting engineering, production, planning, sales and finance on one product model.
- Built the manufacturing master data: ~40,000 materials across 11 material types, 34,000+ BOMs (~1.2M items), ~11,000 routings (41,000+ operations), 88 work centers linked to cost centers, and classification/AVC (16 classes, 92 characteristics) for configurable box culverts.
- Trained the technical office and users across departments on master data and engineering-to-SAP workflows, and remain their first-line support after Go-Live.
- Set the material coding standard and built Excel/VBA and Python generators that produce FG, BOM, routing and quality-inspection data from reference products, loaded in bulk through Migration Cockpit and record by record through OData.
- Delivered 29 automated management reports from S/4HANA OData (sales, production, dispatch, data integrity, QC certificates) as scheduled HTML and Excel dashboards.
- Built a 35-entity OData extraction and local data mirror (~9M rows) that feeds the pricing, engineering and reporting tools without manual exports.

Precast Engineering ERP

- Architect and builder of the Precast Engineering ERP, an inquiry-centred platform for the technical office that runs each order on one inquiry record: BOQ import and item analysis, cost take-off from live SAP BOMs, quotation create and revise, and new FG/BOM/routing/QM creation in SAP, with the design, drawing and submittal engines linked into the same record. Version 1 has run the inquiry-to-quotation workflow daily since Go-Live (1,290+ inquiries).
- Re-engineered it as a tested platform: a Python core with JSON job contracts, an Excel ribbon add-in, a preview-apply-verify-log cycle for every SAP write, and 4,252 automated tests; it already creates cages, routings and production versions directly in SAP.
- Automated design, drawings and BIM for five precast product families (RC pipes, jacking pipes, circular manholes, rectangular manholes and box culverts): design engines that write calculation memos, drawing and BBS generators, and Revit models from the same product data.
- Encoded the company's engineering and product knowledge in software: design rules to ASTM, AASHTO and ACI, SAP-valid product configuration rules and a knowledge base covering 29+ products, so expertise stays with the company rather than with individuals.

BIM Digital Transformation

- Led the BIM transformation of manhole design and drafting: a Revit 2027 product system (C# rule engine and add-in) that turns a design order into native-rebar models, sheets, PDF/DWG/IFC, BBS, BOM and take-off; released v1.0 in Oct 2026 with 1,000+ automated .NET tests.
- Generated a standard drawing library of 6,394 sheets for 1,345 manhole products (3,059 types) from one master data set, each sheet passing an automated drafting-standard check.
- Extended the Revit pipeline to the RC pipe range (67 standard drawing sets), the jacking-pipe range and its steel moulds, every set with 3D views, BBS and automated sheet checks.
- Built parametric Revit families and Dynamo (Python) scripts for precast elements, linking model quantities to Excel and SAP so drawings, BBS and BOMs come from one source.
- Designed the architecture of a multi-tenant precast configurator prototype (Python parametric DXF engine with a Next.js web app; 1,179 tests) that seeded the production manhole tools.

Engineering Systems & AI

- Architect of a multi-agent AI engineering workflow (Claude Code and OpenAI Codex) with model routing, independent cross-model review and safety guards; used to build and maintain the design engines, BIM system, SAP tools and analytics under 17,000+ automated tests across release gates.

Earlier roles

Structural Engineer, United for Telecom Networks Services Company (UTC), Amman, Jordan — **Mar 2023 – Jul 2023**

Design of U-Boot voided slab systems; BOQs and quantity estimates.

Structural Engineer, Al-Dusuqi Engineering Office, Amman, Jordan — **Sep 2022 – Feb 2023**

Analysis and design of RC buildings in ETABS, SAFE and PROKON, verified by hand to ACI; shop drawings and reinforcement detailing.

Structural & Resident Engineer, Al-Majal Engineering Office, Zarqa, Jordan — **Jan 2022 – Aug 2022**

Structural design of building systems, reinforcement schedules and site coordination.

Education

B.Sc. Civil Engineering, The Hashemite University, Jordan, 2022

ABET-accredited programme, GPA 3.20 / 4.00

Training

- Project Management Professional (PMP): in preparation
- Advanced Structural Design Program: reinforced concrete, advanced RC, post-tensioned design, seismic analysis
- BIM Technical Office Engineer Program: parametric modelling, reinforcement detailing, quantity surveying
- Certificates (Google Drive)
- Portfolio (Google Drive)

Languages

Arabic (native)

English (professional working proficiency)

Engineering software

SAP2000 and its API: Finite-element models built and run from code: shells for manholes, a frame on Winkler springs for box culverts, ring frames and 3D solids for RC and jacking pipes, with results read back into the memos.

Eriksson Pipe: Cross-check of the RC pipe design engine; the comparison agreed within 0.04%.

Eriksson Culvert: Precast box culvert design and checking.

ETABS, SAFE, PROKON: Analysis and design of RC buildings, slabs and foundations, verified by hand to ACI.

Revit 2027 and its C# API: Product systems that generate families, native rebar, views, sheets, BBS and BOM: manholes, RC pipes, jacking pipes and their steel moulds.

Dynamo, AutoCAD: Parametric modelling and DWG issue alongside the Revit sheets.

SAP S/4HANA Cloud: Material master, BOMs, routings, work centers and classification through Fiori, Migration Cockpit, Mass Maintenance and OData APIs.

Python, Excel, VBA: Design engines, memo and catalogue generators, the engineering ERP and the SAP data tools, behind 17,000+ automated tests.

Power BI, SQLite: A local SAP data mirror and code-generated production dashboards.

Claude Code, OpenAI Codex: Multi-agent development with cross-model review and test gates.

Tools

SAP: S/4HANA Cloud Public Edition (MM/PP), Fiori, Migration Cockpit, Mass Maintenance, OData APIs, classification and AVC. **Structural analysis:** SAP2000 including API automation, Eriksson Culvert, Eriksson Pipe, ETABS, SAFE, PROKON. **BIM / CAD:** Revit 2027 (families, native rebar, C# API add-ins), Dynamo, AutoCAD, IFC / DWG export. **Data & automation:** Python, Excel (LET, LAMBDA, XLOOKUP), VBA, SQL/SQLite, Power BI, HTML reporting. **AI engineering:** Claude Code, OpenAI Codex, multi-agent orchestration, automated verification gates. **Standards:** ASTM C76, C478, C497, C1433, AASHTO LRFD, ACI 318, MOMRA, BS 5911.